

- (a)** Complete Table 3.1 to compare the structure of *V. cholerae* and an epithelial cell from the human intestine.

Use a cross (X) to indicate the absence of the structure.

Table 3.1

cell structure	<i>V. cholerae</i>	epithelial cell from the human intestine
cell wall	✓	x
cell surface membrane		
ribosomes		
large permanent vacuole		
organelles surrounded by a double membrane		

(b) Describe the methods that can be used to prevent the spread of cholera.

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[5]

- (c) The World Health Organization has suggested that people with HIV/AIDS take a longer time to recover from cholera and are at an increased risk of death from cholera.

The human immunodeficiency virus (HIV) only infects certain types of cell. These cells have CD4 receptor proteins in their cell surface membrane. Helper T-lymphocytes have CD4 receptor proteins.

- (i) Fig. 3.1 is a diagram of HIV showing the glycoprotein gp120.

This glycoprotein is embedded in a membrane envelope which surrounds the viral protein coat.

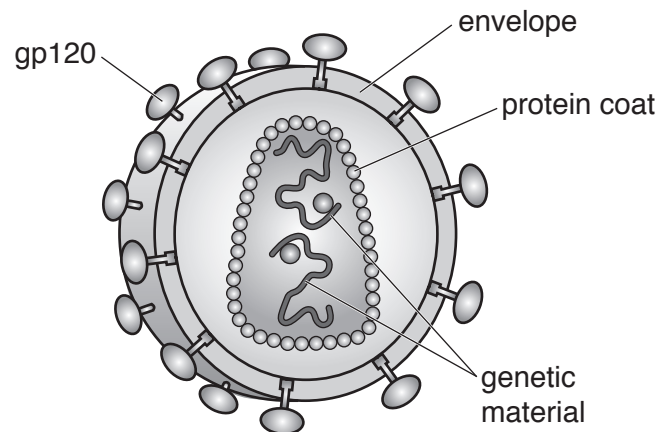


Fig. 3.1

The glycoprotein gp120 is important in allowing HIV to only infect certain types of cell.

Suggest the role of gp120.

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- (ii) People with HIV/AIDS have a low helper T-lymphocyte count.

Explain how a low helper T-lymphocyte count could reduce the body's ability to produce antibodies against *V. cholerae*.

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[Total: 13]